Modeling Life Expectancy and Country Development Indicators

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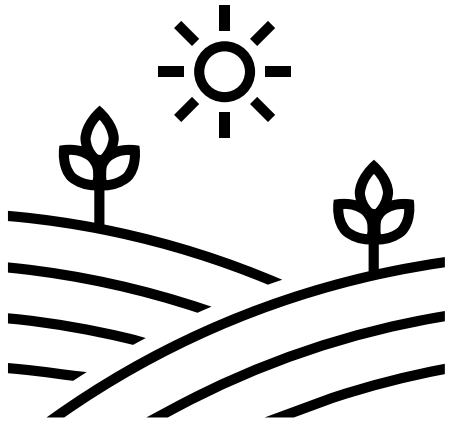
DTSC3010

University of North Texas

Fall 2024



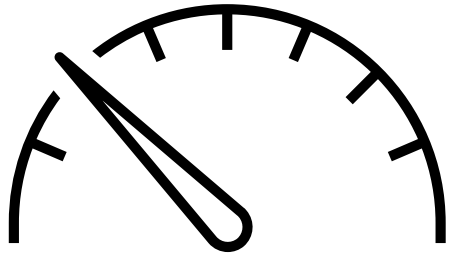
Introduction



Millenium Development Goals
(2000-2015)



Sustainable Development Goals
(2015-2030)



Gauge progre
further



Research Question

How do different development indicators, such as BMI, immunization rates, schooling, and GDP, contribute to a country's life expectancy and development status?



Dataset

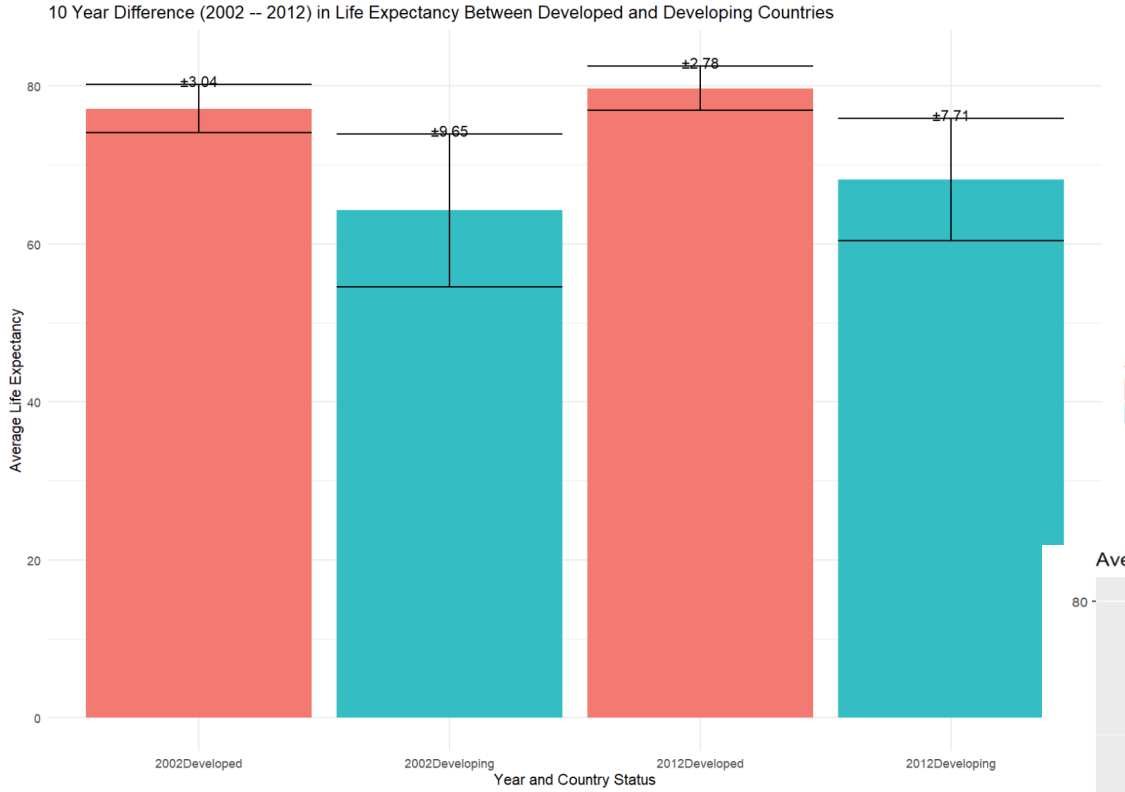
Classification	Mortality and Burden of Disease	Disease	Nutrition	Economic and Intellectual	Miscellaneous
Country Name	Life expectancy	Measles immunization rates	BMI	GDP per capita	Alcohol consumption
Year	Infant deaths	Hepatitis B immunization rates	Thinness (age 10-19)	Schooling level	
Region	Under five deaths	Polio immunization rates	Thinness (age 5-9)	Economy status	
	Adult mortality	Diphtheria immunization rates			
		HIV incidents			

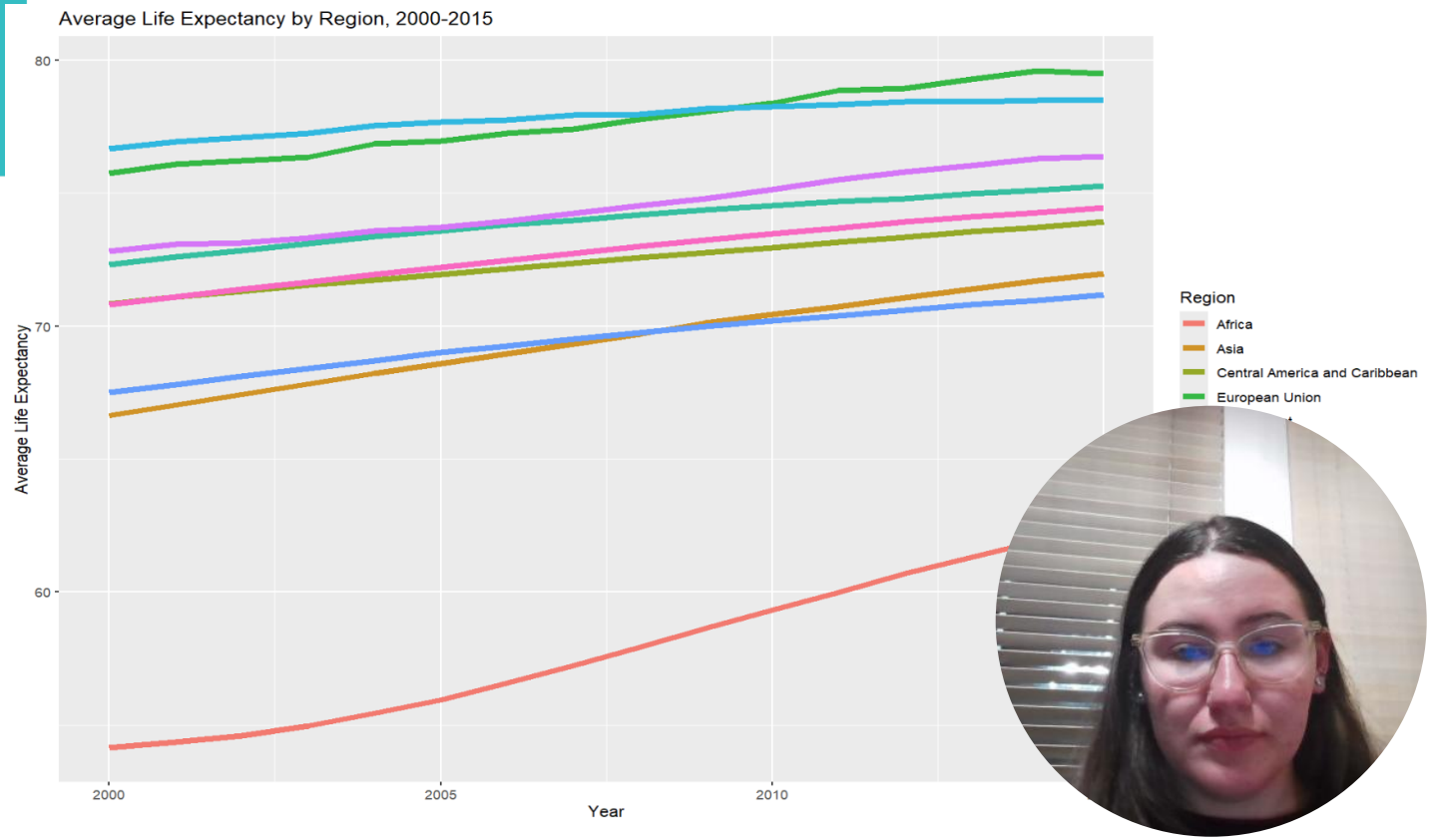


Statistical Summary

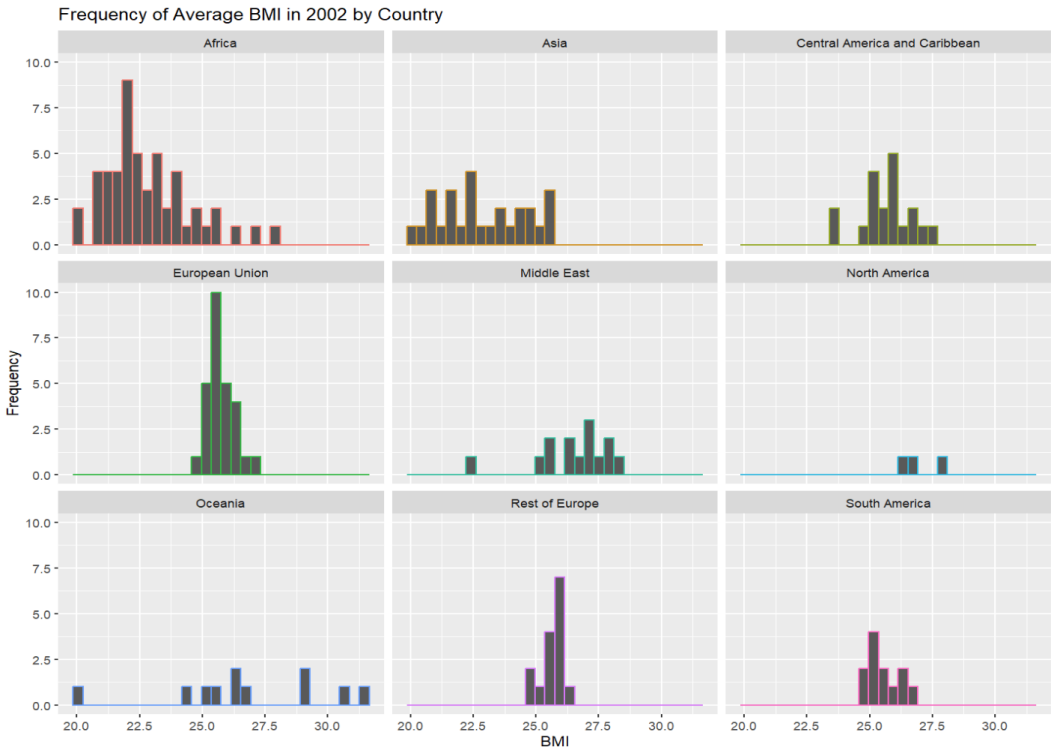
Unimodal, Right-Skewed	Unimodal, Left-Skewed	Bimodal	Unimodal, symmetric
Infant deaths	Polio immunization rates	Measles immunization rates	BMI
Under five deaths	Hepatitis B immunization rates		Schooling
Adult mortality	Diphtheria immunization rates		
Alcohol consumption			
HIV incidents			
Thinness age 10-19			
Thinness age 5-9			
GDP per capita			

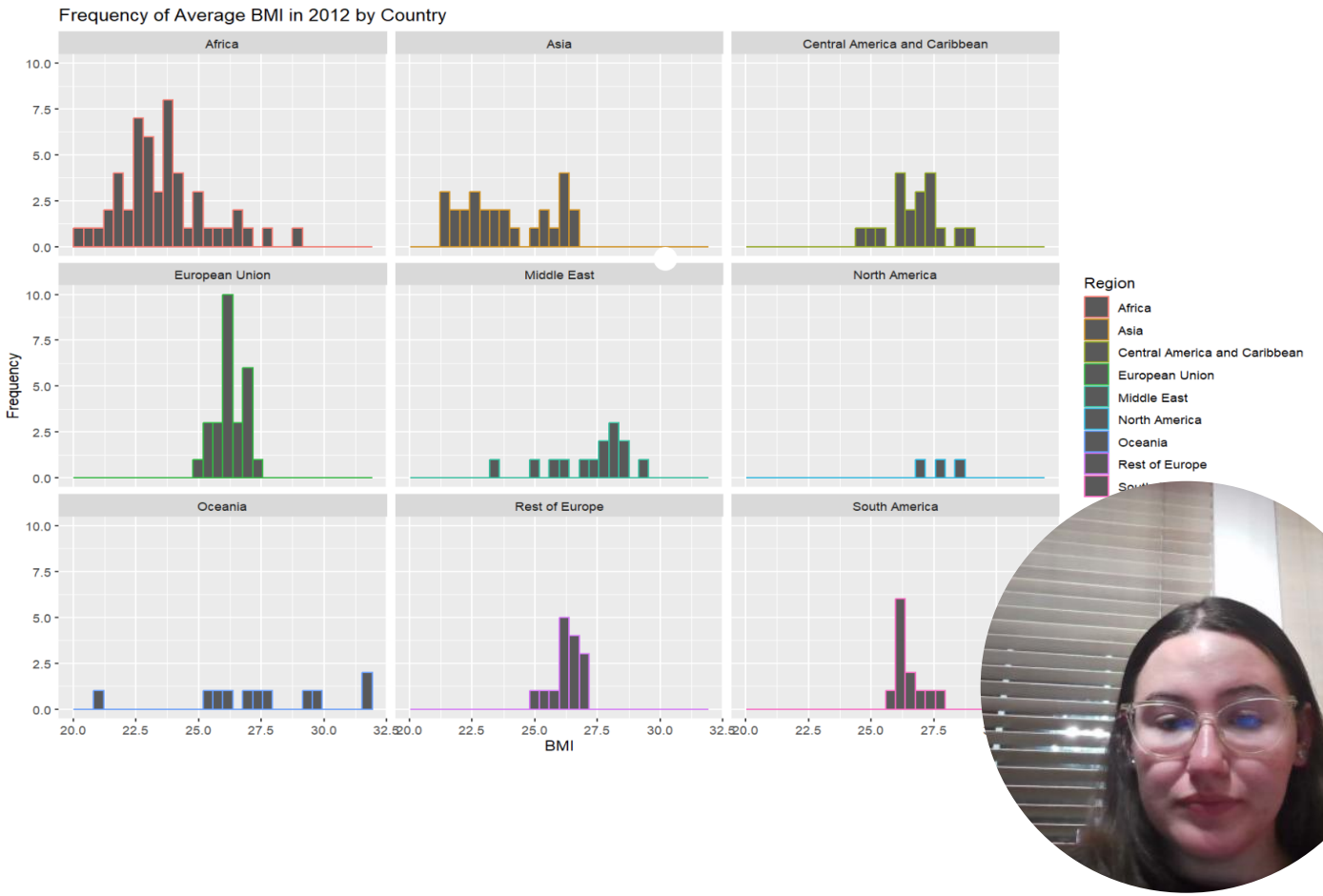


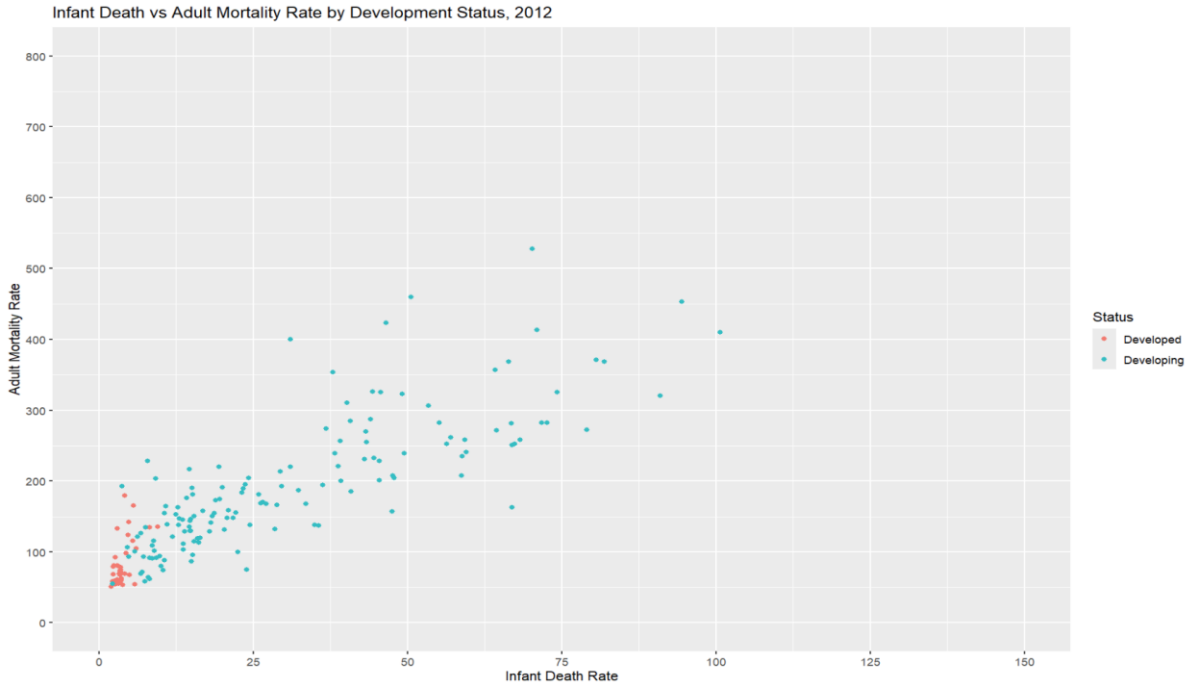

10 Year Difference (2002 -- 2012) in Life Expectancy Between Developed and Developing Countries
±3.04
±9.65
±2.78
±7.71
80
60
40
20
0
Average Life Expectancy
2002Developed
2002Developing
2012Developed
2012Developing
Year and Country Status
Status
Developed
Developing

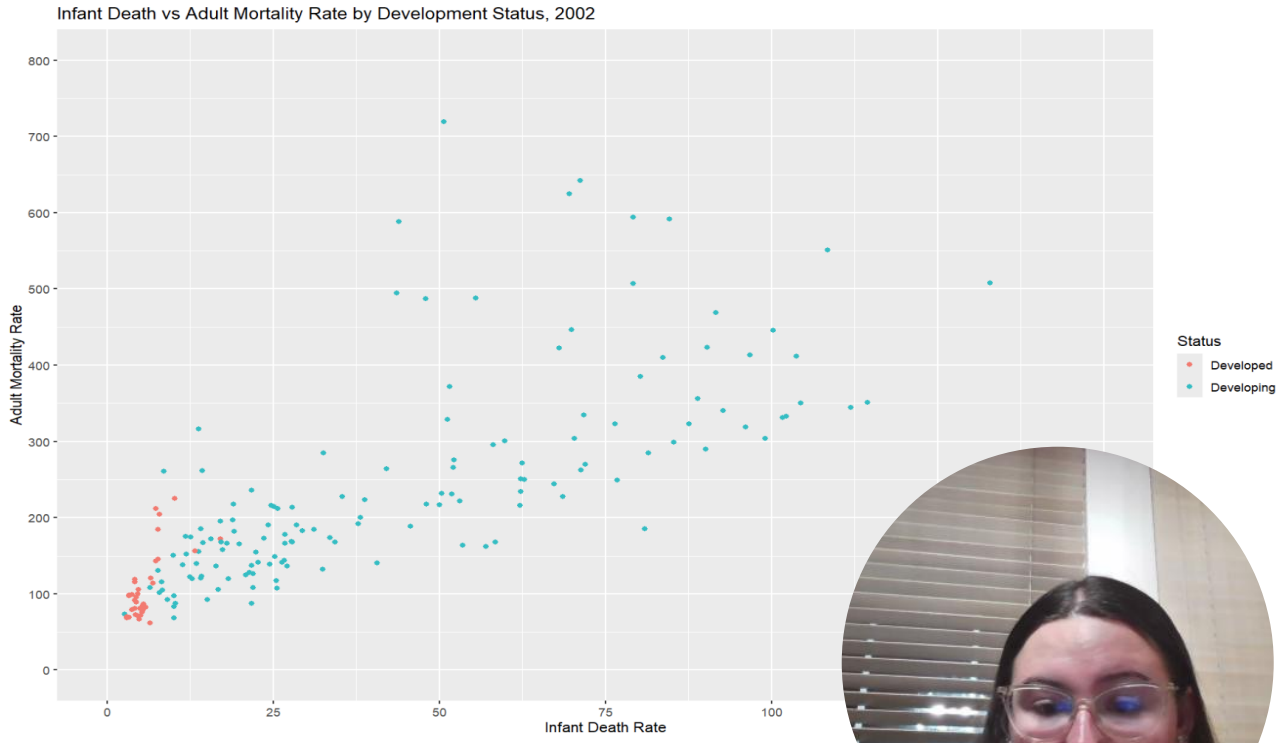

Average Life Expectancy by Region, 2000-2015
80
70
60
Average Life Expectancy
2000
2005
2010
Year
Region
Africa
Asia
Central America and Caribbean
European Union

BMI


Frequency of Average BMI in 2002 by Country
Africa
Asia
Central America and Caribbean
European Union
Middle East
North America
Oceania
Rest of Europe
South America
Frequency
BMI


Frequency of Average BMI in 2012 by Country
Africa
Asia
Central America and Caribbean
European Union
Middle East
North America
Oceania
Rest of Europe
South America
Frequency
BMI
Region
Africa
Asia
Central America and Caribbean
European Union
Middle East
North America
Oceania
Rest of Europe


Infant Death vs Adult Mortality Rate by Development Status, 2012
Adult Mortality Rate
Infant Death Rate
0
100
200
300
400
500
600
700
800
0
25
50
75
100
125
150
Status
Developed
Developing


Infant Death vs Adult Mortality Rate by Development Status, 2002
Adult Mortality Rate
Infant Death Rate
0
100
200
300
400
500
600
700
800
0
25
50
75
100
Status
Developed
Developing



T-Test

```
## Statistical Test 1 - T Test

```{r}

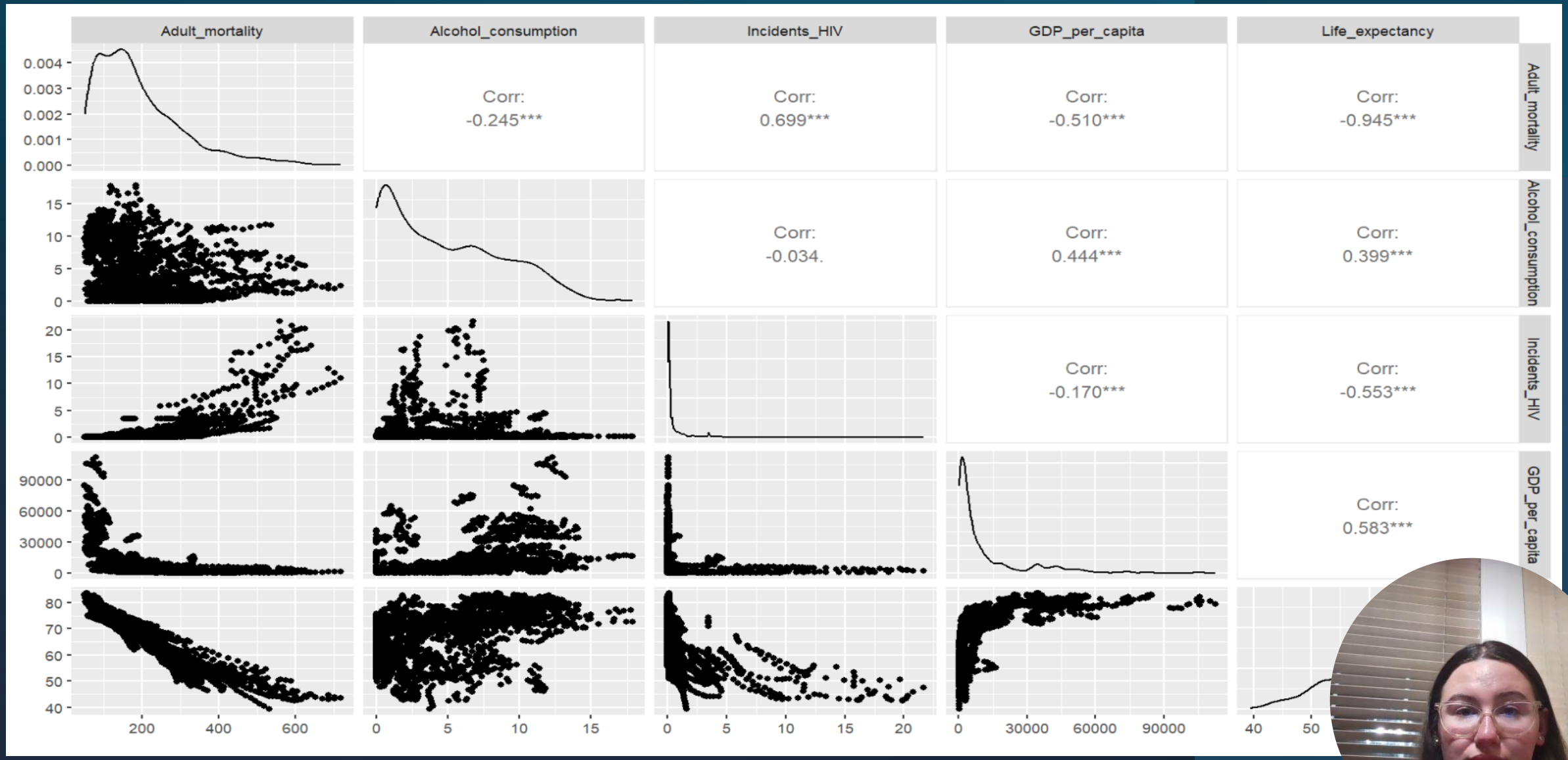
t.test(Life_expectancy~Status, data = life_expectancy)

```
```

```
        Welch Two Sample t-test

data:  Life_expectancy by Status
t = 53.685, df = 2623.8, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Developed and group Developing is not equal to
0
95 percent confidence interval:
 11.71972 12.60832
sample estimates:
 mean in group Developed mean in group Developing
                78.50574                 66.34173
```




Adult_mortality
Alcohol_consumption
Incidents_HIV
GDP_per_capita
Life_expectancy
Corr:
-0.245***
Corr:
0.699***
Corr:
-0.510***
Corr:
-0.945***
Corr:
-0.034.
Corr:
0.444***
Corr:
0.399***
Corr:
-0.170***
Corr:
-0.553***
Corr:
0.583***
0.004
0.003
0.002
0.001
0.000
15
10
5
0
20
15
10
5
0
90000
60000
30000
0
80
70
60
50
40
200
400
600
0
5
10
15
0
5
10
15
20
0
30000
60000
90000
40
50

Linear Model – Life Expectancy

- Strong, statistically significant relationship with schooling
- 61.13% variance explained by predictors
- MAE = 4.614408

```
Call:
lm(formula = Life_expectancy ~ GDP_per_capita + Schooling + BMI +
    Alcohol_consumption, data = life_expectancy)

Coefficients:
        (Intercept)       GDP_per_capita            Schooling                  BMI  Alcohol_consumption
         33.1232108            0.0001442            1.4538087            0.9546478           -0.1918141


Call:
lm(formula = Life_expectancy ~ GDP_per_capita + Schooling + BMI +
    Alcohol_consumption, data = life_expectancy)

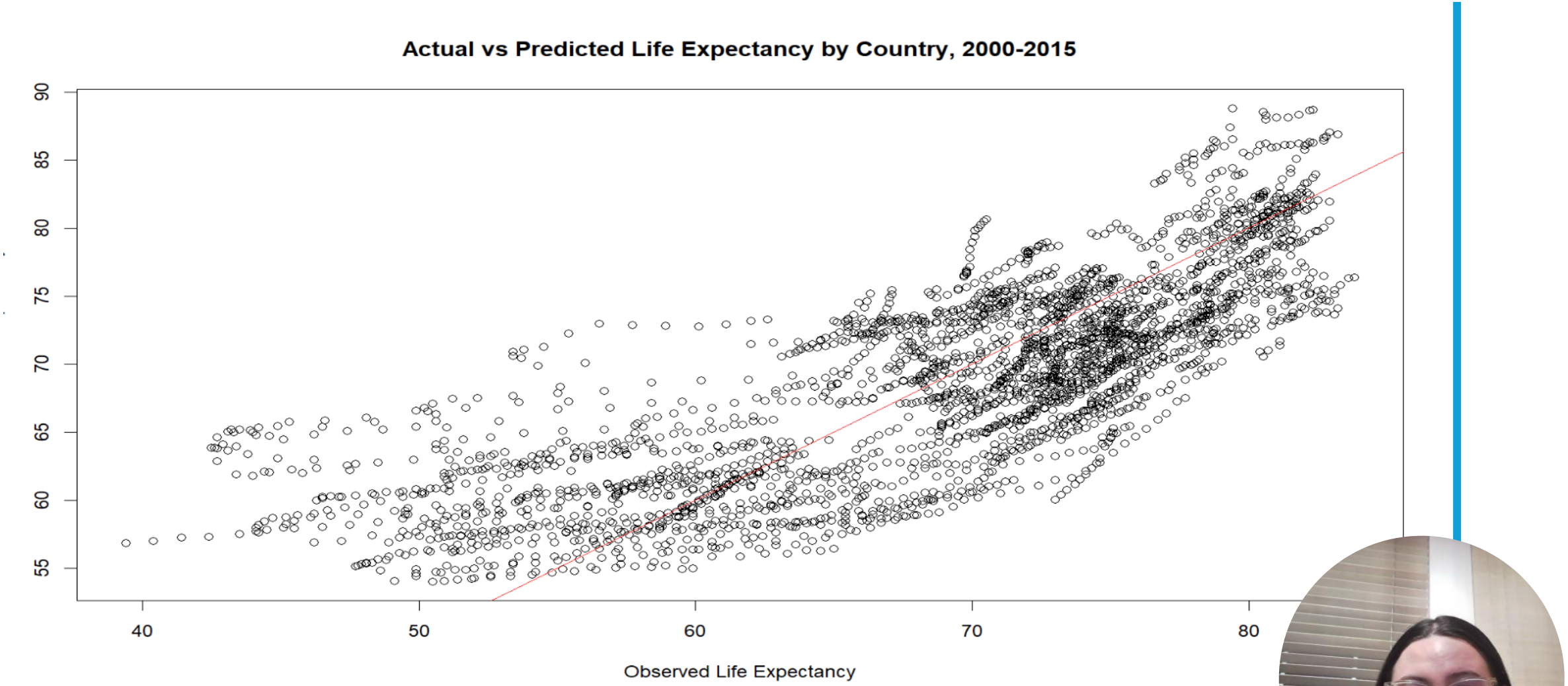
Residuals:
     Min       1Q   Median       3Q      Max
-21.9911  -3.4823   0.6203   4.3137  12.9962

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        3.312e+01  1.449e+00  22.861  < 2e-16 ***
GDP_per_capita     1.442e-04  8.031e-06  17.957  < 2e-16 ***
Schooling          1.454e+00  5.965e-02  24.373  < 2e-16 ***
BMI                9.546e-01  6.580e-02  14.509  < 2e-16 ***
Alcohol_consumption -1.918e-01  3.580e-02  -5.358  9.1e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.868 on 2859 degrees of freedom
Multiple R-squared:  0.6113,    Adjusted R-squared:  0.6108
F-statistic:  1124 on 4 and 2859 DF,  p-value: < 2.2e-16

[1] 4.614408
```




Actual vs Predicted Life Expectancy by Country, 2000-2015
90
85
80
75
70
65
60
55
40
50
60
70
80
Observed Life Expectancy

GLM Model – Development Status

```
Call:
glm(formula = Economy_status_Developed ~ Hepatitis_B + Measles +
    Polio + Diphtheria + Thinness_ten_nineteen_years + Thinness_five_nine_years +
    GDP_per_capita, family = binomial(link = "logit"), data = train_glm)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -1.404e+01  1.760e+00  -7.975 1.52e-15 ***
Hepatitis_B                -3.490e-02  1.280e-02  -2.727  0.00639 **
Measles                     5.002e-02  9.028e-03   5.541 3.01e-08 ***
Polio                      -7.464e-02  3.094e-02  -2.412  0.01584 *
Diphtheria                  2.179e-01  3.571e-02   6.101 1.05e-09 ***
Thinness_ten_nineteen_years -1.417e+00  1.525e-01  -9.294  < 2e-16 ***
Thinness_five_nine_years    5.478e-02  1.161e-01   0.472  0.63710
GDP_per_capita              8.768e-05  6.656e-06  13.174  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2322.47  on 2290  degrees of freedom
Residual deviance:  793.22  on 2283  degrees of freedom
AIC: 809.22

Number of Fisher Scoring iterations: 8

          Actual
Predicted   0   1
        0 437  20
        1  13 103
```

80/20 split

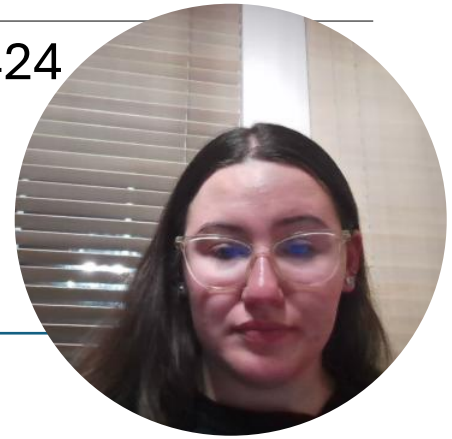
Predictors explain large amount of variance

Childhood nutrition and immunization rates



GLM Model – Confusion Matrix

Metric	Formula	Result
Precision: correct positive predictions	True Positive / (True Positive + False Positive)	103 / 116 = 0.8879
Recall: complete positive predictions	True Positive / (True Positive + False Negative)	103 / 123 = 0.8374
Specificity: correct negative predictions	True Negative / (True Negative + False Positive)	437 / 450 = 0.9711
Accuracy: correctly classified	(True Positive + True Negative) / (True Positive + True Negative + False Positive + False Negative)	540 / 573 = 0.9424



Random Forest Model – Life Expectancy

	Overall <dbl>
Adult_mortality	100.00000
Infant_deaths	85.77459
Incidents_HIV	14.98653
GDP_per_capita	0.00000

60/40 split + 5fold CV
Relative importance
Decreasing infant deaths
RMSE = 0.7987866



SVM Poly Model – Development Status

```
Confusion Matrix and Statistics

             Reference
Prediction    Developed Developing
  Developed         156          24
  Developing         40         640

               Accuracy : 0.9256
                 95% CI : (0.906, 0.9422)
    No Information Rate : 0.7721
    P-Value [Acc > NIR] : < 2e-16

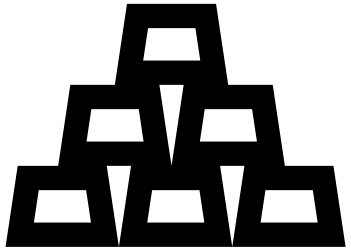
                  Kappa : 0.7823

 Mcnemar's Test P-Value : 0.06079

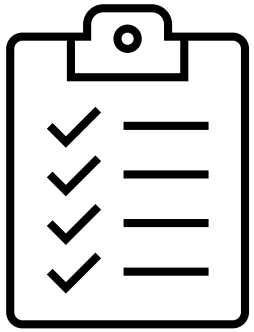
            Sensitivity : 0.7959
            Specificity : 0.9639
         Pos Pred Value : 0.8667
         Neg Pred Value : 0.9412
             Prevalence : 0.2279
         Detection Rate : 0.1814
   Detection Prevalence : 0.2093
      Balanced Accuracy : 0.8799

       'Positive' Class : Developed
```

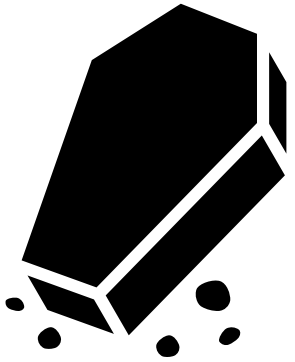




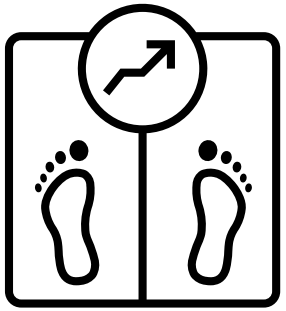
GDP not enough



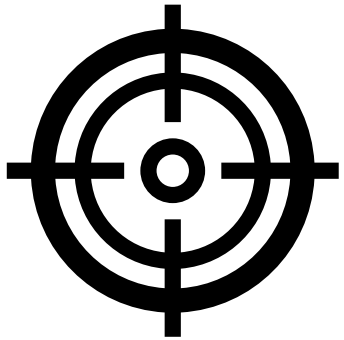
Work on immunization, childhood nutrition, infant death rate, and education



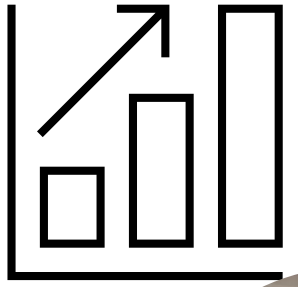
Decrease in adult mortality and infant death rates



Increase in BMI

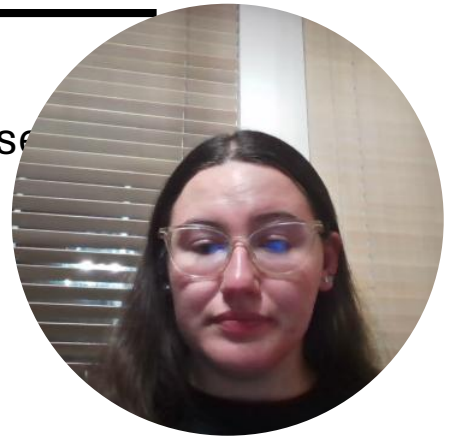


Wide disparity in Africa



Overall increase
expectancy

Conclusions



Error / Future

- Ambiguity in definitions
- Further understanding of models
- Trying different indicators/combinations of indicators
- More current data ex. COVID

References

In addition to class resources:

Plotting actual vs predicted values: How to Plot Predicted Values in R (With Examples)

Interpreting GLM output: How to Interpret glm Output in R (With Example)

Feature Importance Charts: Interpreting Random Forest Classification Results - GeeksforGeeks

Adding standard deviation bars to charts: Plot mean and standard deviation using ggplot2 in R - GeeksforGeeks

Correlograms: Correlogram – from Data to Viz

Dataset: https://www.kaggle.com/datasets/lashagoch/life-expectancy-who-updated

•This webpage also has the links to the WHO landing pages for each indicator

Resources for Introduction (MDG + SDG):

MDG Monitor. (n.d.). Retrieved from Millenium Development Goals: https://www.mdgmonitor.org/millennium-development-goals/

Nations, U. (n.d.). *Sustainable Development Goals*. Retrieved from Goals: https://sdgs.un.org/goals

